

4	A 3.00 g copper coin at 20.0 °C drops 50.0 m to the ground.		
	(a)	The copper is said to possess internal energy. Explain what is meant by the internal energy.	
			
			
			
			[2]
		Solution The internal energy of a system is the sum of its <u>kinetic energy due to the random motion</u> of its molecules and <u>potential energy due to intermolecular forces of attraction</u> .	B1 B1
	(b)	The coin does not undergo a change in volume after it landed on the ground. Determine the gain in temperature of the coin given that the specific heat capacity of copper is 385 Jkg ⁻¹ K ⁻¹ . Assuming that 10.0 % of the change in gravitational potential energy of the coin goes to increasing the internal energy of the coin.	
		gain in temperature = K	[2]
		Solution By principle of conservation of energy, decrease in G.P.E. → increase in internal energy $0.10mg\Delta h = mc\Delta\theta$ $\Delta\theta = \frac{0.10mg\Delta h}{mc} = \frac{0.10(9.81)(50.0)}{385}$ $= 0.13 \text{ K}$	M1 A1
	(c)	The first law of thermodynamics for a system can be expressed as $\Delta U = q + w$ where ΔU is the increase in internal energy of the system, q is the heat supplied to the system and w is the work done on the system. Use the words positive , negative and zero to complete Table 4.1 for the three terms in the equation for each of the processes shown. You may use each word once, more than once, or not at all.	

			<table><thead><tr><th>Process</th><th>ΔU</th><th>q</th><th>w</th></tr></thead><tbody><tr><td>Copper coin drops and lands on the ground</td><td></td><td></td><td></td></tr></tbody></table>	Process	ΔU	q	w	Copper coin drops and lands on the ground				
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Table 4.1					[2]							
		Solution ΔU: positive q: positive (B1 for both ΔU and q) w: zero – B1 one mark here As the coin's temperature rises, its average molecular kinetic energy increases. Thus, the increase in internal energy of the system ΔU is positive. As the coin does not undergo any change in volume, the work done on the system w is zero. By the First Law, since ΔU is positive and $w = 0$, the heat supplied to the system q must be positive.			B1 B1							